

Evaluation Criteria — L1- inception-requirements- extractor-agent

Quality Gates (must pass)

Criterion	Threshold	Method
IDs sequential	FR-01, FR-02... / NFR-01... / CON-01... / ASM-01... / GAP-01...	Automated: pattern check
No duplicate IDs	All IDs unique across categories	Automated: uniqueness check
Priorities assigned	Every FR and NFR has MoSCoW priority	Automated: field check
Citations present	Every FR and NFR has citation.source_reference	Automated: field check
No empty required fields	id, title, description populated	Automated: null check
No invented features	Every requirement traces to input text	LLM-as-Judge: faithfulness
Confidence calibrated	Explicit = 0.9+, inferred = 0.7-0.8	Automated: range check

Evaluation Scores (LLM-as-Judge)

Evaluator	Threshold	Direction
Faithfulness	≥ 0.90	Every requirement traces to input phrase
Hallucination	≤ 0.10	No invented capabilities or features
Consistency	≥ 0.90	Requirements don't contradict each other
Relevance	≥ 0.85	Requirements are actionable, not vague
Reasoning quality	≥ 0.80	Categorisation decisions explained
Citation completeness	≥ 0.95	Every item cites exact input phrase
Completeness	≥ 0.85	Input's capabilities adequately captured

Requirements Quality Rubric

Dimension	Score 9-10	Score 7-8	Score 5-6	Score < 5
-----------	------------	-----------	-----------	-----------

Dimension	Score 9-10	Score 7-8	Score 5-6	Score < 5
Extraction	All stated features captured + implied ones flagged	Most stated captured, 1-2 missed	Several features missed	Major gaps
Categorisation	FRs, NFRs, constraints, assumptions correctly separated	Mostly correct, 1 miscategorised	Several miscategorised	Categories confused
Priority	MoSCoW aligned with input language strength	Mostly correct priorities	Several wrong priorities	Random
Gaps & Assumptions	Genuine gaps identified with impact + question	Most gaps real, 1 over-flagged	Under-flagged (missing obvious gaps)	No gaps identified
Measurability	NFRs have quantifiable criteria	Most measurable, 1 vague	Several vague NFRs	No measurable criteria

Reflection Checklist

The agent must self-verify before delivering:

- ☐ All explicitly stated capabilities have a corresponding FR
- ☐ NFRs are measurable (time, percentage, uptime — not "fast" or "secure")
- ☐ Constraints are fixed decisions, not capabilities (technology choices, not features)
- ☐ Assumptions are genuinely inferred, not fabricated
- ☐ Gaps are actionable — each has an impact statement and suggested question
- ☐ No requirement contradicts another
- ☐ Confidence scores match: explicit statements = 0.9+, inferences = 0.7-0.8
- ☐ Execution summary includes category counts and coverage assessment

Reflection Process (mandatory)

The agent **MUST** perform reflection before delivering output:

- Generate** initial output following processing rules
- Log** [REFLECTING] Checking output against evaluation criteria
- Check** every item in the Reflection Checklist above
- Identify** gaps, errors, inconsistencies, or missed items
- Log** each finding: [REFLECTING] Found: <description>
- Fix** each issue — amend the output silently
- Log** each resolution: [REFLECTING] Resolved: <what was fixed>
- Deliver** only the final corrected output

The reflection findings and resolutions should appear in the execution_summary (what reflection found and changed) but the interim output must never be shown.